

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2017

Mathematics

Paper 2

Higher Level

Time: 2 hours, 30 minutes

300 marks

School stamp

Running total	
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For examiner			
Question	Mark	Question	Mark
1		11	
2			
3			
4			
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 11 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if you do not show all necessary work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

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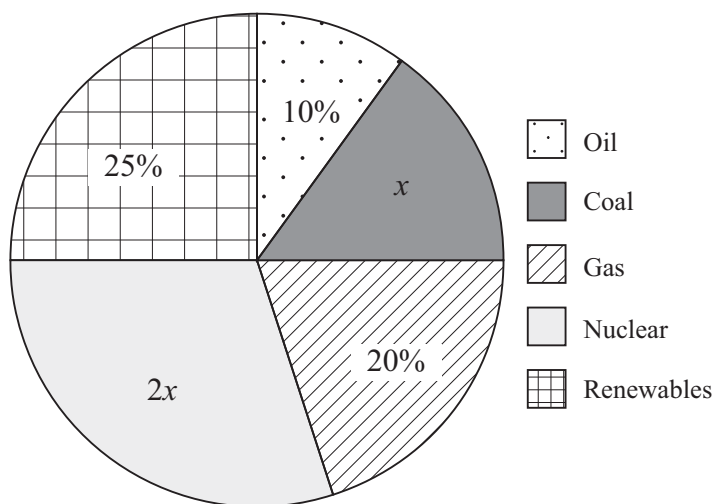
Question 1

(Suggested maximum time: 10 minutes)

The pie chart opposite shows the sources of energy production in the EU in 2016.

- (a)** What type of data is displayed?
Put a tick (✓) in the correct box.

Numerical Discrete	☐
Numerical Continuous	☐
Categorical Nominal	☐
Categorical Ordinal	☐



- (b)** What percentage of energy was produced from coal in 2016?

[illegible]

- (c) Calculate the measure of the angle in the pie chart that represents the percentage of energy produced from nuclear in 2016.

[illegible]

- (d) 75 tonnes of energy was produced from renewables in 2016. How much energy was produced from oil in 2016?

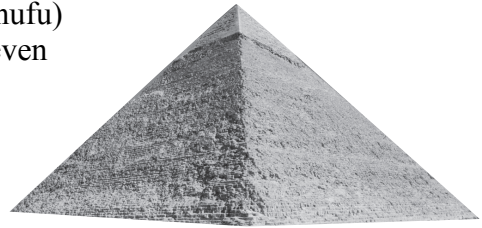
[illegible]

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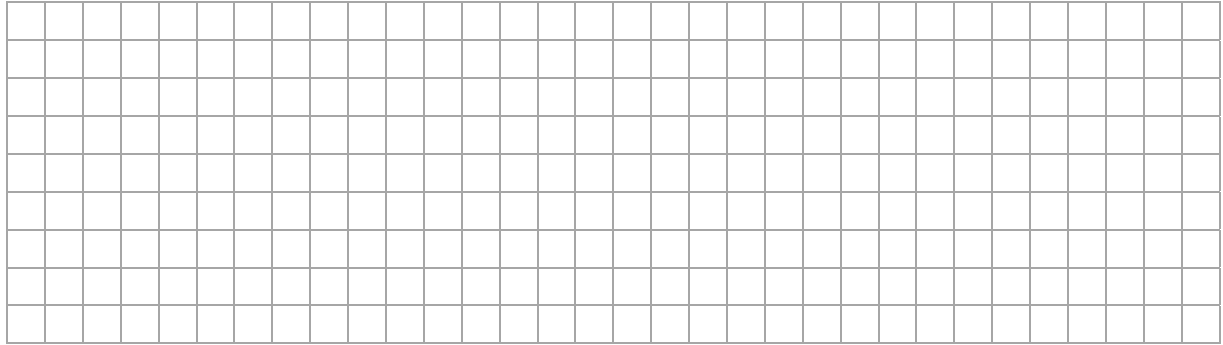
Question 2**(Suggested maximum time: 15 minutes)**

The Great Pyramid of Giza (also known as the Pyramid of Khufu) on the banks of the River Nile in Egypt is the oldest of the Seven Wonders of the Ancient World.

The pyramid consists of four equilateral triangular sides sitting on a square base.

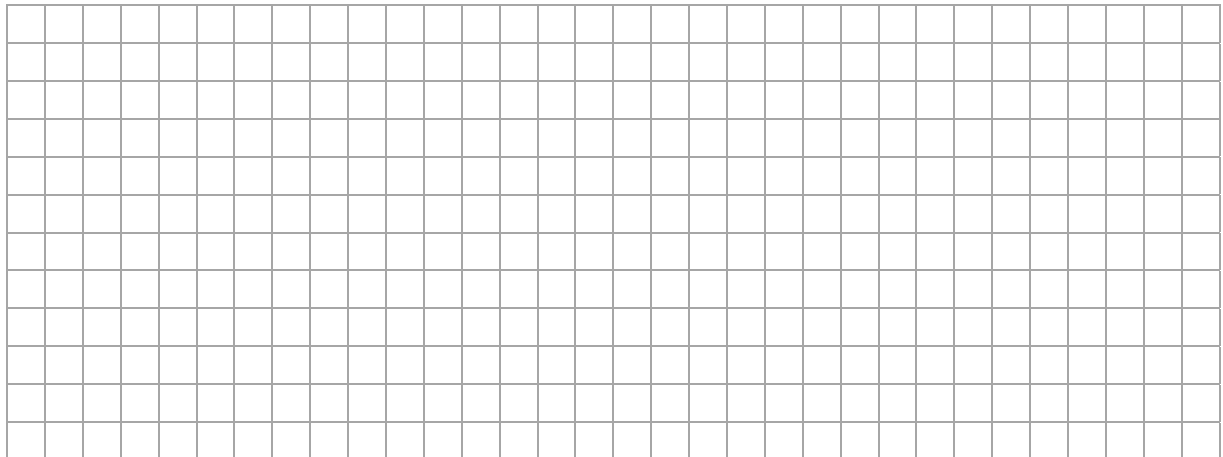


- (a) The base of the pyramid has an area of $52\,900\text{ m}^2$. Find the length of one side of its base.

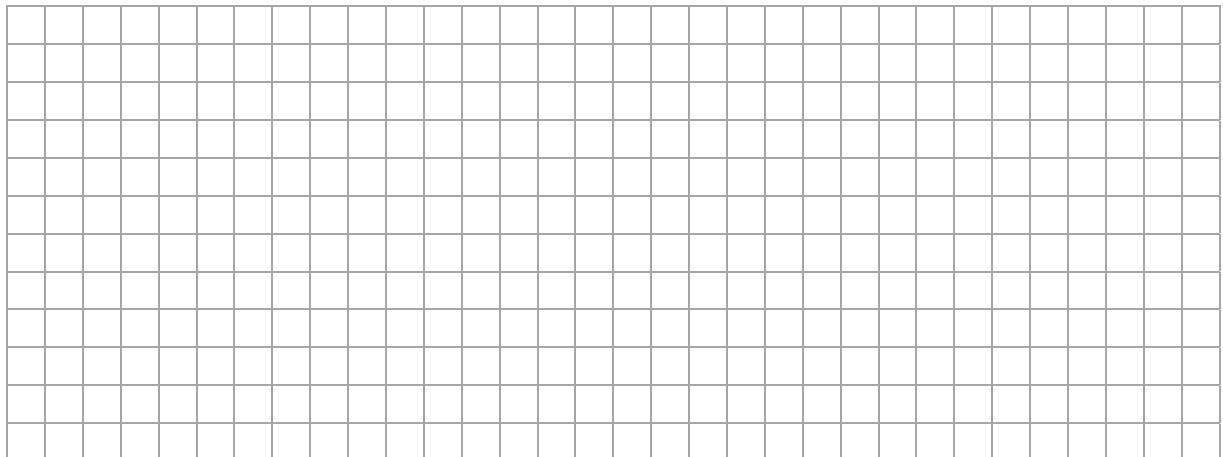


- (b) When it was built, the triangular faces of the pyramid were covered in polished limestone.

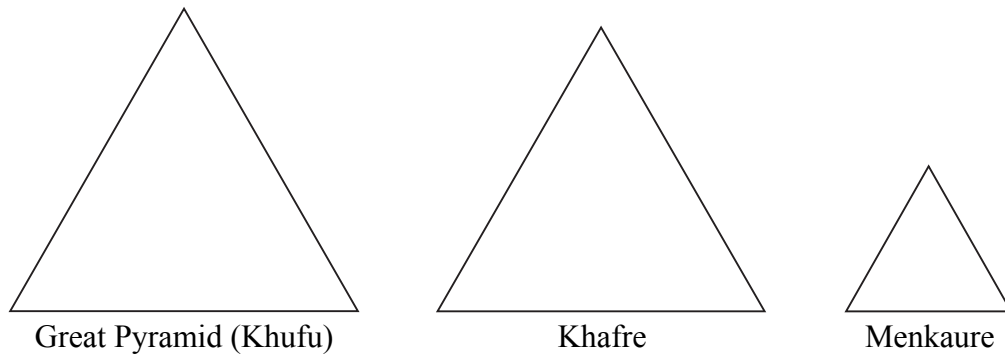
- (i) Find the height of one of the triangular faces.
Give your answer correct to the nearest metre.



- (ii) Find the surface area of the polished limestone that would have been required to cover the pyramid.



- (c) There are two smaller pyramids, Khafre and Menkaure, near the Great Pyramid. The triangular faces of all three pyramids are similar, as shown in the diagram below.



- (i) Find the percentage decrease in the length of a side of Khafre, whose sides measure 216 m compared to that of the Great Pyramid (Khufu). Give your answer correct to one decimal place.

[illegible]

- (ii) The percentage decrease in the length of a side of Menkaure compared to that of Khafre is 49.5%. Calculate the length of the perimeter of the base of Menkaure. Give your answer correct to the nearest metre.

[illegible]

- (iii) Does the surface area of the faces of the pyramids decrease at the same rate? Give a reason for your answer.

Answer:

Reason:

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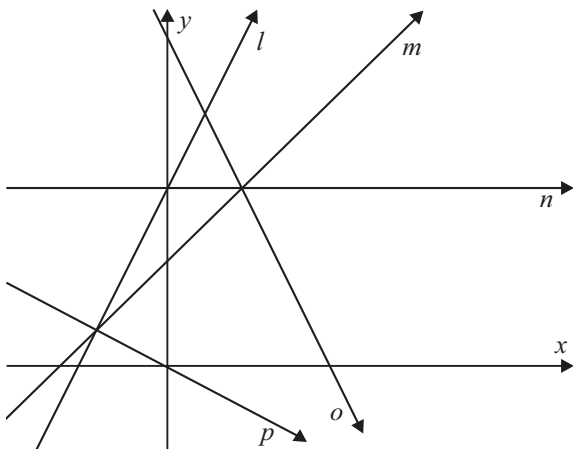
Question 3

(Suggested maximum time: 10 minutes)

- (a)** The co-ordinate diagram shows the lines l , m , n , o and p .

The table shows the equation of each line.

Write the letters l , m , n , o and p into the table to match each line to its equation.



Equation	Line
$x - y + 3 = 0$	
$x = -2y$	
$y - 5 = 0$	
$y = 2x + 5$	
$2x + y = 9$	

[illegible]

- (b)** Complete the following sentences. Write one of the letters l , m , n , o and p in each box.

- (i) Lines and have negative slopes.

- (ii) Lines and are perpendicular to each other.

- (iii) Lines and intersect at the point $(0, 5)$.

- (c) The lines m , n and o intersect at the same point.
Find the co-ordinates of this point of intersection.

A full-page view of a blank sheet of graph paper. The grid consists of thin, light gray horizontal and vertical lines forming small squares across the entire page. There are no margins, text, or other markings on the paper.

Question 4

(Suggested maximum time: 10 minutes)

(a) Construct a right-angled triangle ABC , where:

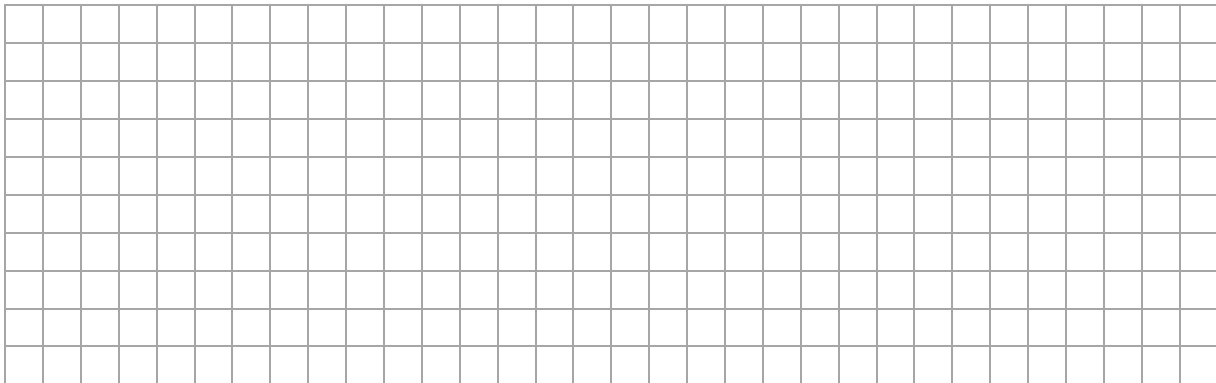
$|AB| = 6 \text{ cm}$
 $|\angle CAB| = 90^\circ$
 $|BC| = 8 \text{ cm}.$



(b) On your diagram, measure the angle $\angle ABC$.
Give your answer correct to the nearest degree.

$|\angle ABC| = \underline{\hspace{2cm}}$

(c) Using trigonometry, verify your answer to part (b) above.



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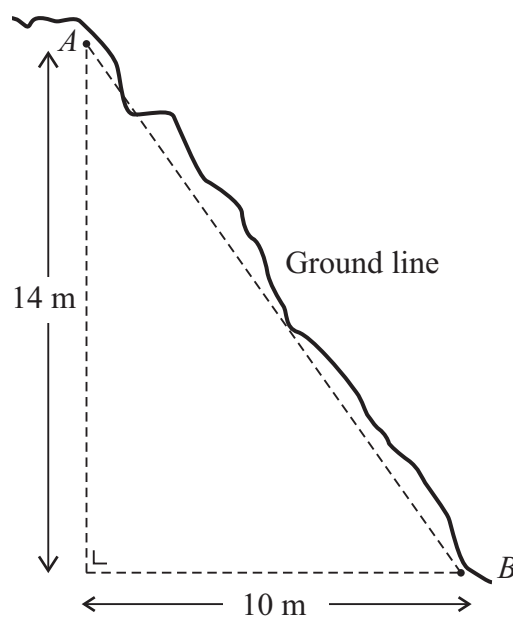
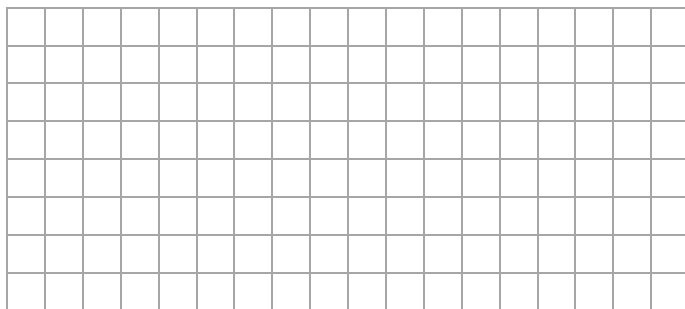
Question 5

(Suggested maximum time: 15 minutes)

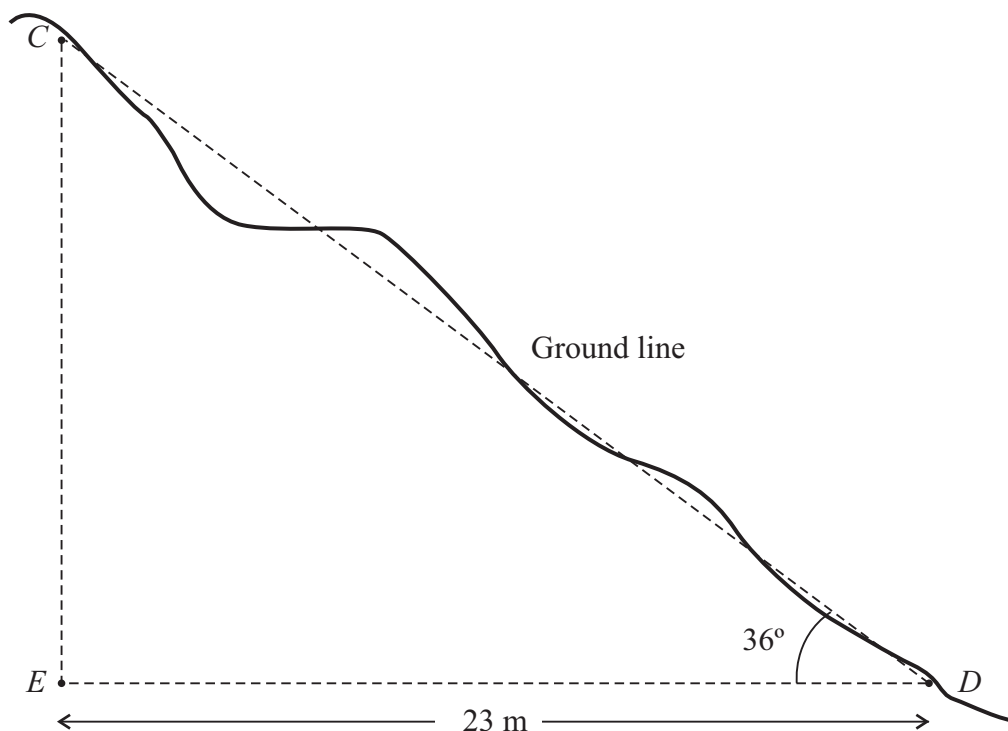
Landslides occur when the stability of a slope of ground is compromised. This can happen after there has been heavy rainfall.

The sketch shows the incline (slope) of a section of ground on the side of a mountain prior to a landslide.

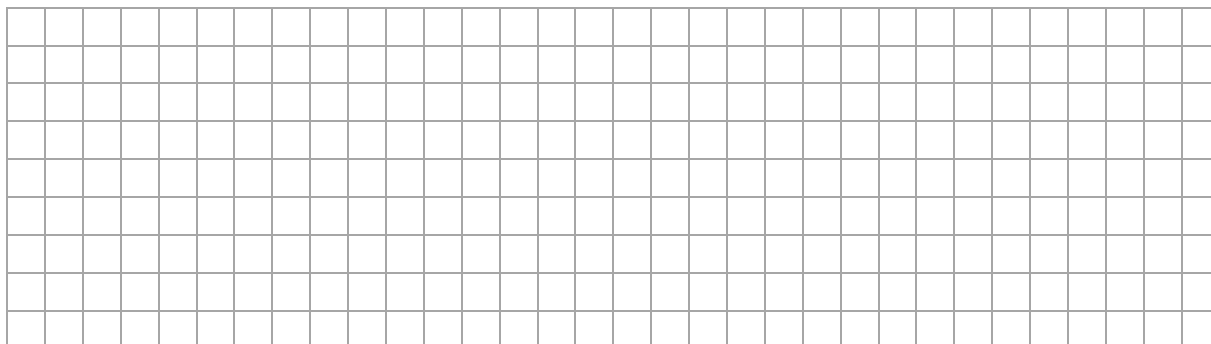
- (a) Find the incline (slope) of the land in the sketch between the points A and B .



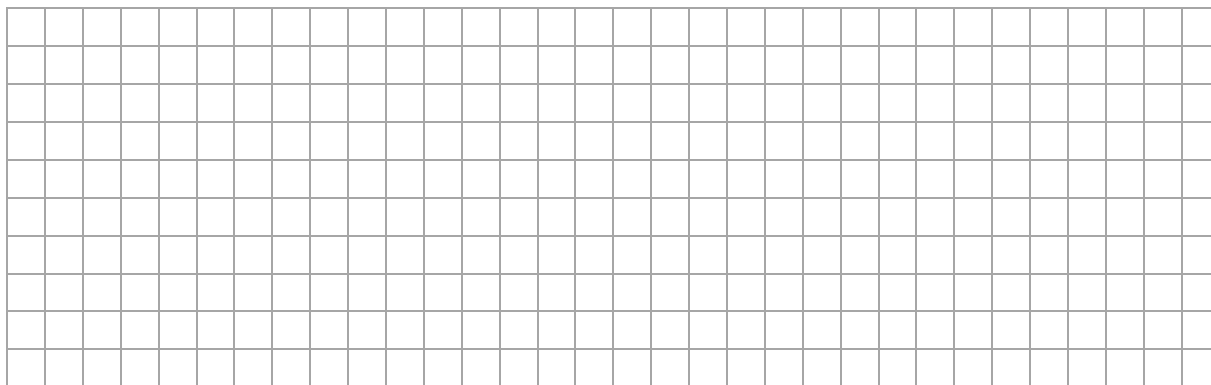
- (b) The diagram below shows a sketch of the incline after a landslide. $|\angle EDC| = 36^\circ$.



- (i) Find the incline (slope) of the ground shown in the sketch between the points C and D .



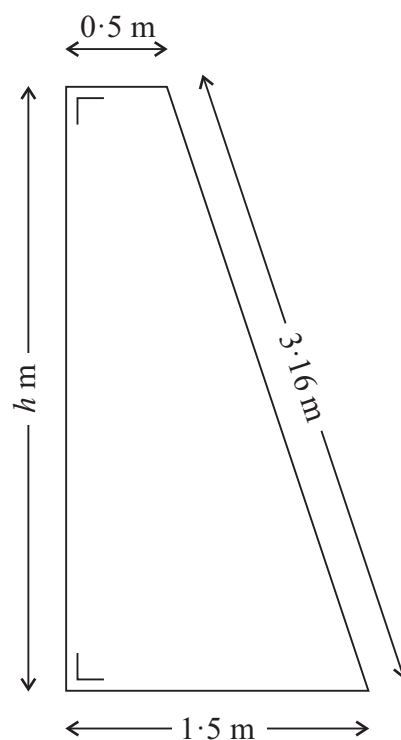
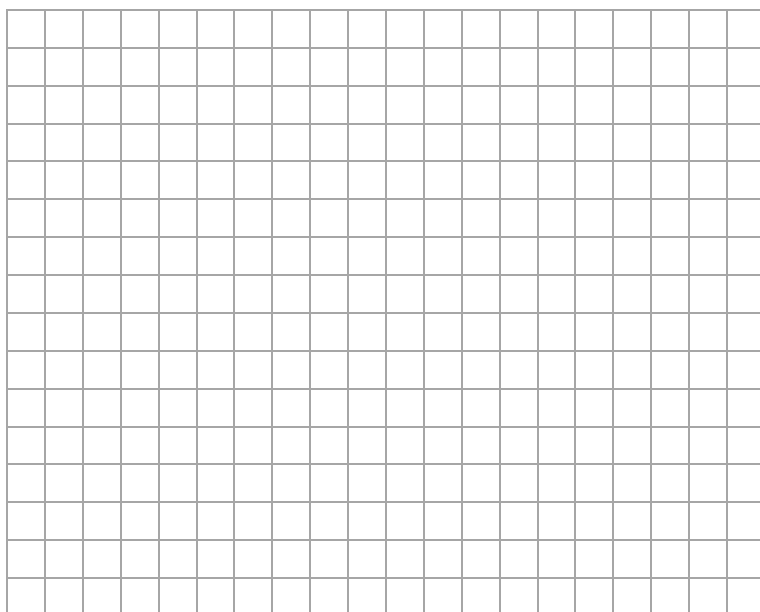
- (ii) Hence, find $|CE|$, giving your answer correct to the nearest metre.



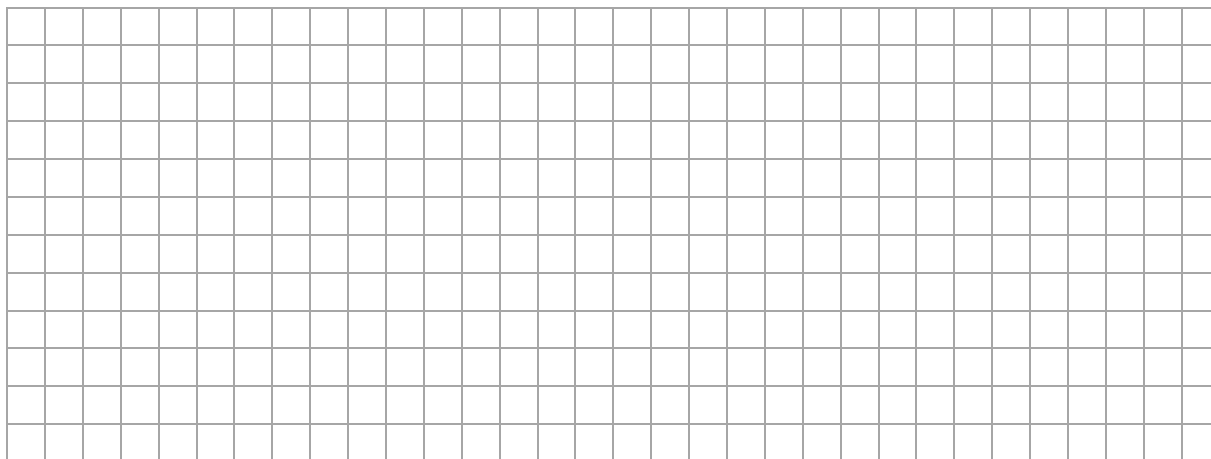
- (c) The National Roads Authority has decided to construct a concrete retaining wall, 80 m long, to protect a section of motorway from future landslides.

The cross-section of the retaining wall, of height h , is shown.

- (i) Find h , giving your answer correct to the nearest metre.



- (ii) Given that the wall is 80 m long, find the volume of concrete required to construct the wall.



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(Suggested maximum time: 25 minutes)

Junior School						Senior School							
				8	6	0	3	5	7	8	9		
8	6	5	4	3	2	1	1	4	5	5	5	7	9
		8	6	4	3	1	2	3	4	5	8	9	
						2	3	4	6	7			
						3	4	0					
						4	0						
						5	9						

Key: | 2 | 2 means 22 minutes

- [illegible]

- [illegible]

- [illegible]

- 11

- | Time (minutes) | 0 – 10 | 10 – 20 | 20 – 30 | 30 – 40 | 40 – 50 | 50 – 60 |
|----------------|--------|---------|---------|---------|---------|---------|
| Frequency | | | | | | |

[illegible]

- (f) Use **mid-interval values** of the data in the grouped frequency table to estimate the **mean** time taken to travel to school by the students in the sample.
Give your answer correct to the nearest minute.

- (g) Display the data from the grouped frequency table on a histogram.

A full-page sheet of white graph paper featuring a light gray grid. The grid consists of small, equal-sized squares arranged in a continuous pattern across the entire surface. There are no margins, text, or other markings on the page.

- (h) The students felt that the wording of Q1 in the survey could have been improved, because some students didn't take the survey seriously when filling in their answers as shown.

<i>Green Schools Project Survey</i>	
Q1: How did you travel to school this morning?	<i>by Jet Pack</i>
Q2: How long did it take you to get to school this morning?	<i>25 minutes</i>

Rewrite Q1 of the survey in a more suitable form.

[illegible]

Question 7

(Suggested maximum time: 20 minutes)

- (a)** An electric appliance store sells three different brands of television, all in various screen sizes and features, as shown.

Televisions		
Brand	Features	Screen Size
Zony Hamsung Fillips	Smart TV Full HD	24 inches
		32 inches
		40 inches
		49 inches
		55 inches
		65 inches

- (i)** Calculate how many different televisions can be purchased from this store.

[illegible]

- (ii) A customer purchased a television from the store. Assuming all brands and types of televisions are equally likely to be purchased, find the probability that this person purchased a Zony 32 inch smart TV.

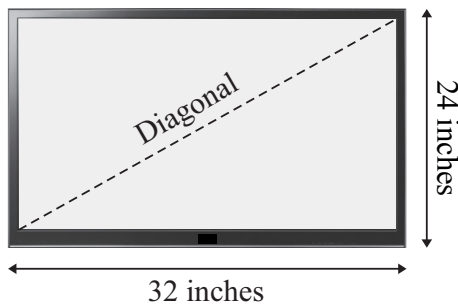
[illegible]

- (iii) A customer is chosen at random from those who bought a Fillips TV. Find the probability that this person purchased a smart TV.

[illegible]

- (iv) Television screen sizes are measured according to the length of their diagonal. A television on special offer has a screen 24 inches in height and 32 inches in width.

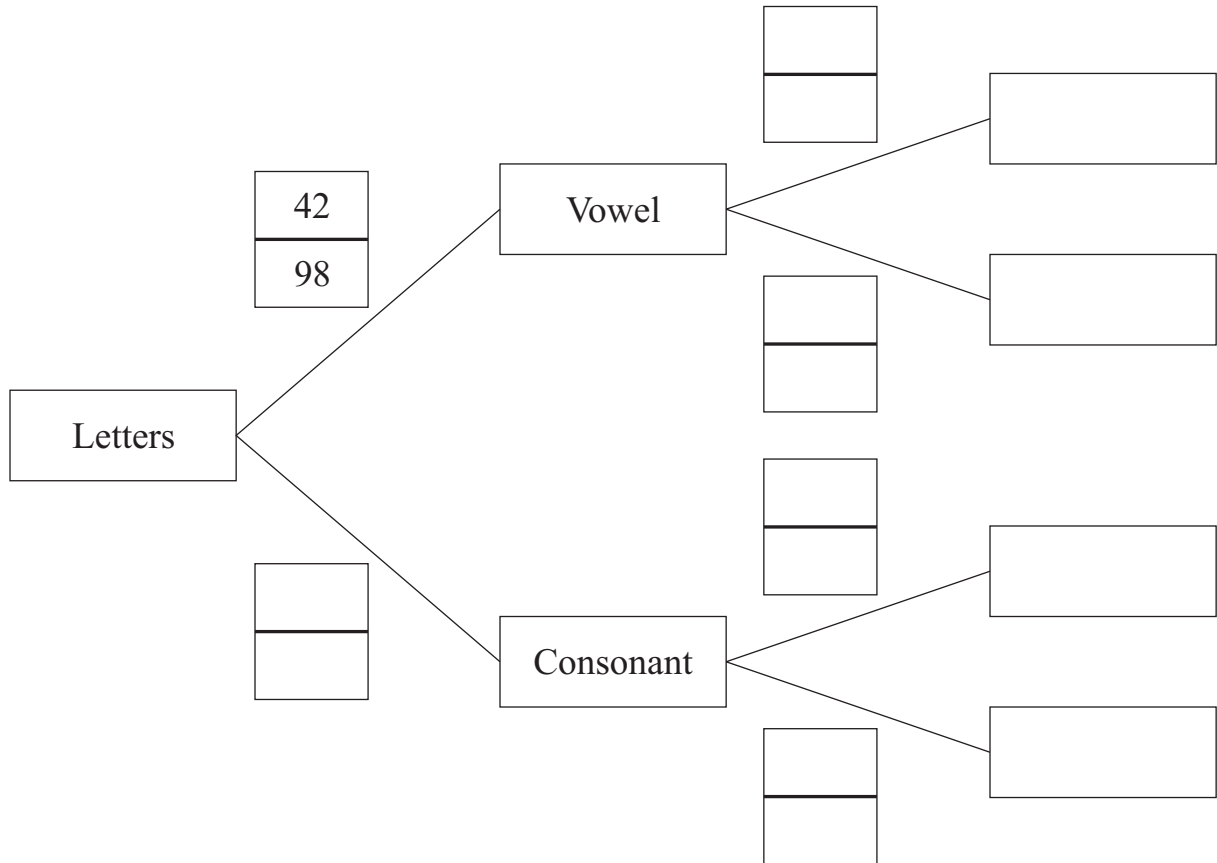
Write down the screen size of the television.
Give a reason for your answer.



Answer:

Reason:

- (b) The English-language edition of Scrabble contains 100 tiles, 98 of which have a single letter printed on them. There are 42 vowels and 56 consonants. To make a game more difficult, the two blank tiles are removed. The board game starts with each player picking tiles from a bag containing the remaining 98 tiles. When a tile is picked from the bag, it is not replaced.
- (i) The first player picks two tiles from the bag. Complete the tree diagram below to show all possible outcomes and fill in the boxes in the diagram which represent the probability that vowels or consonants will be picked.



- (ii)** Find the probability that the tiles selected are either both consonants or both vowels.

[illegible]

- (iii) Find the probability that at least one of the tiles selected is a vowel.

[illegible]

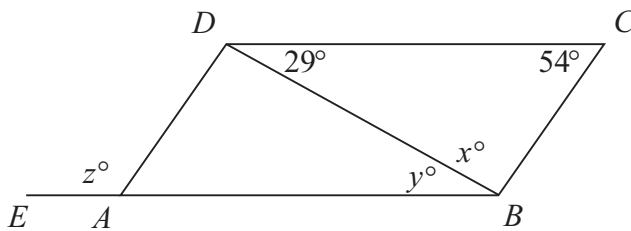
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Question 8

(Suggested maximum time: 10 minutes)

- (a)** $ABCD$ is a parallelogram with the side AB extended to E to form an external angle.

- (i) Find the measure of each of the angles marked x , y and z .
Give a reason for your answer in each case.



Angle:	Reason:
$x =$	
$y =$	
$z =$	

- (ii) What type of triangle does DBC represent? Give a reason for your answer.

[illegible]

- (iii) State whether triangles DBC and ABD are **congruent**.
Give a reason for your answer.

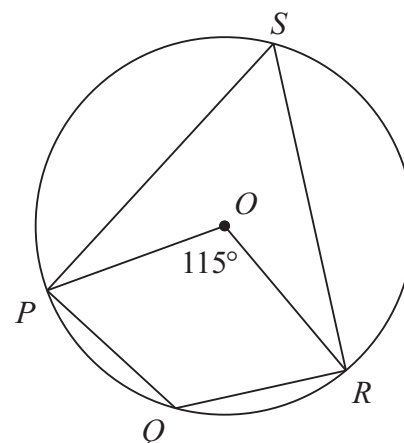
[illegible]

- (b)** P, Q, R and S are points on a circle with centre O . $|\angle ROP| = 115^\circ$, as shown.

- (i) Find $|\angle RSP|$.

[illegible]

- (ii) Find $|\angle PQR|$.

[illegible]

Question 9 (Suggested maximum time: 5 minutes)

Question 9 (Suggested maximum time: 5 minutes)

Prove that the angle at the centre of a circle standing on a given arc is twice the angle at any point of the circle standing on the same arc.

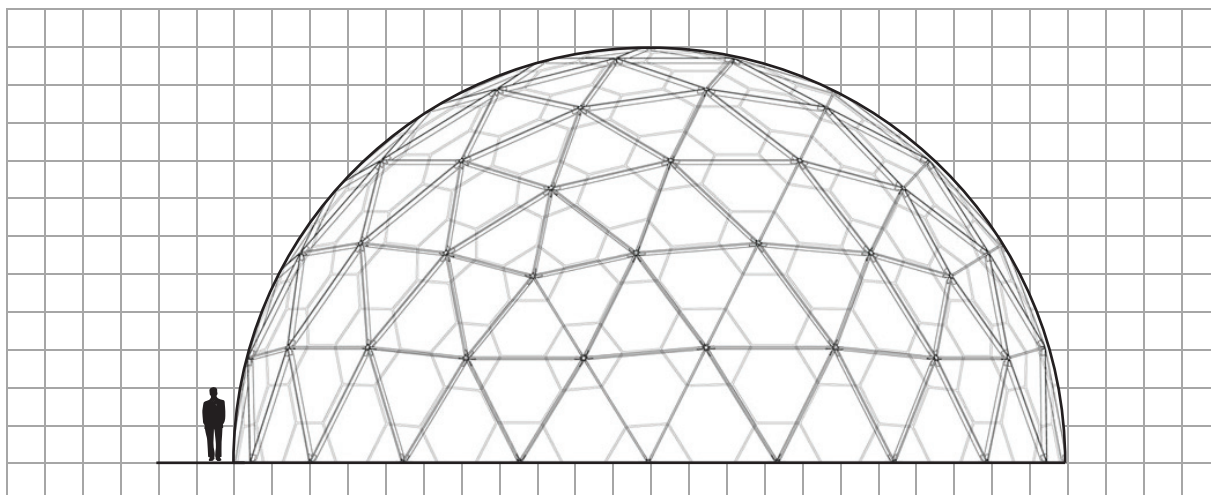
Diagram:

<i>Given:</i>	
<i>To Prove:</i>	
<i>Construction:</i>	
<i>Proof:</i>	

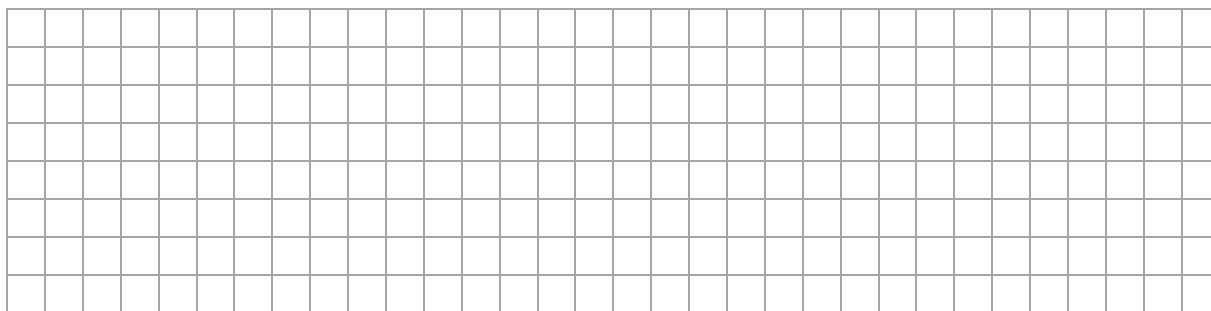
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Question 10**(Suggested maximum time: 10 minutes)**

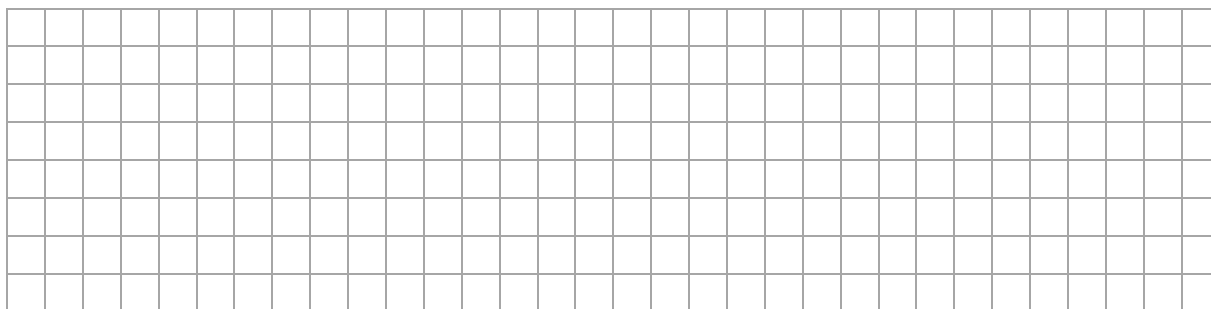
The diagram below is a scale drawing of a dome which is in the shape of a hemisphere. The figure of a man, who is 1.8 m tall, standing beside the dome, allows the scale of the drawing to be estimated.



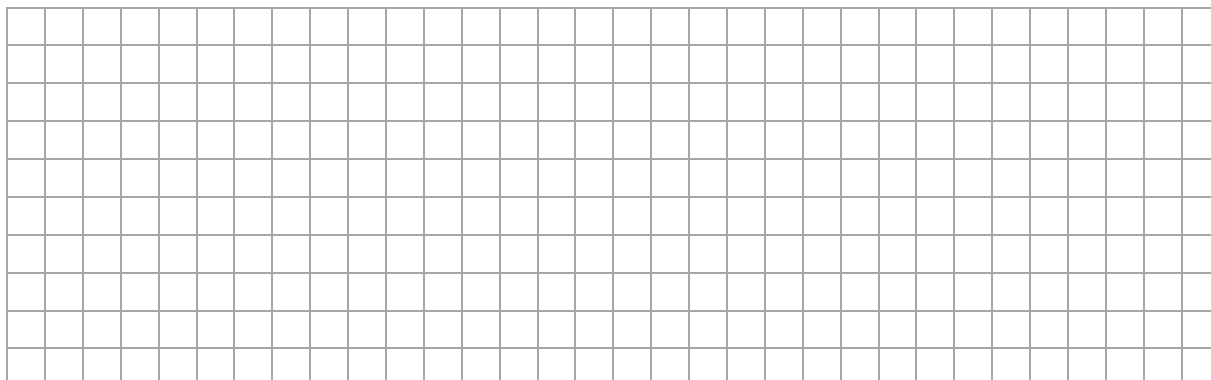
- (a)** Estimate the radius of the dome.



- (b)** Using your answer to part **(a)**, find the curved surface area of the dome.
Give your answer in m^2 , correct to one decimal place.



- (c)** The volume of the dome is 2063.14 m^3 . Find the radius of the dome.
Give your answer in metres, correct to two decimal places.



- (d)** Find the percentage difference between the actual radius and the estimated radius, correct to one decimal place.

- (e) Find the area of the floor of the dome, in terms of π .

[illegible]

Question 11**(Suggested maximum time: 10 minutes)**

The equation of the line l is $3x - y - 1 = 0$.

- (a) Find the co-ordinates of the points where l cuts the axes.

l cuts the x -axis at (,)	l cuts the y -axis at (,)
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- (b) Find the length of the line segment between the two points found in part (a).

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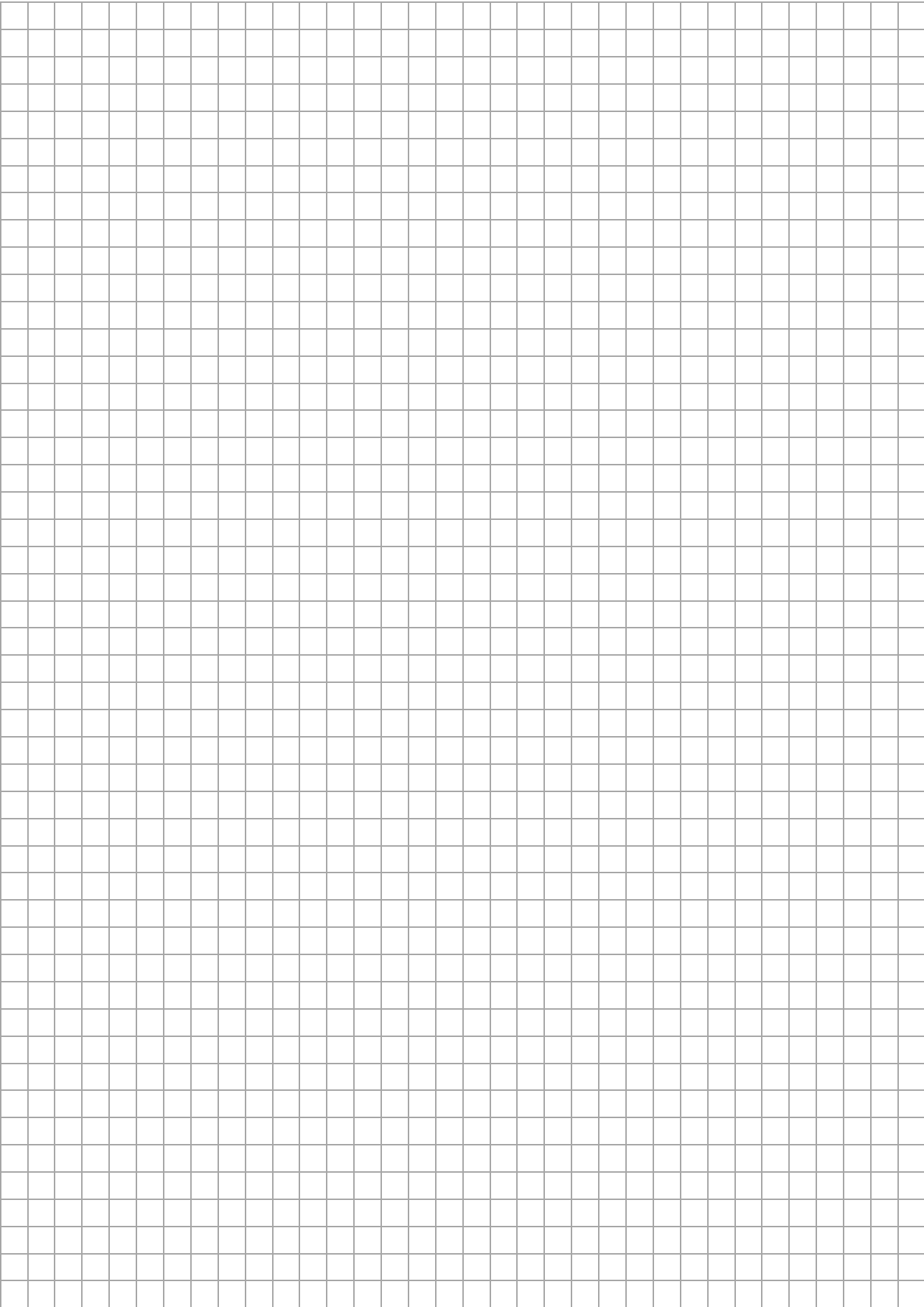
- (c) Find the slope of the line l .

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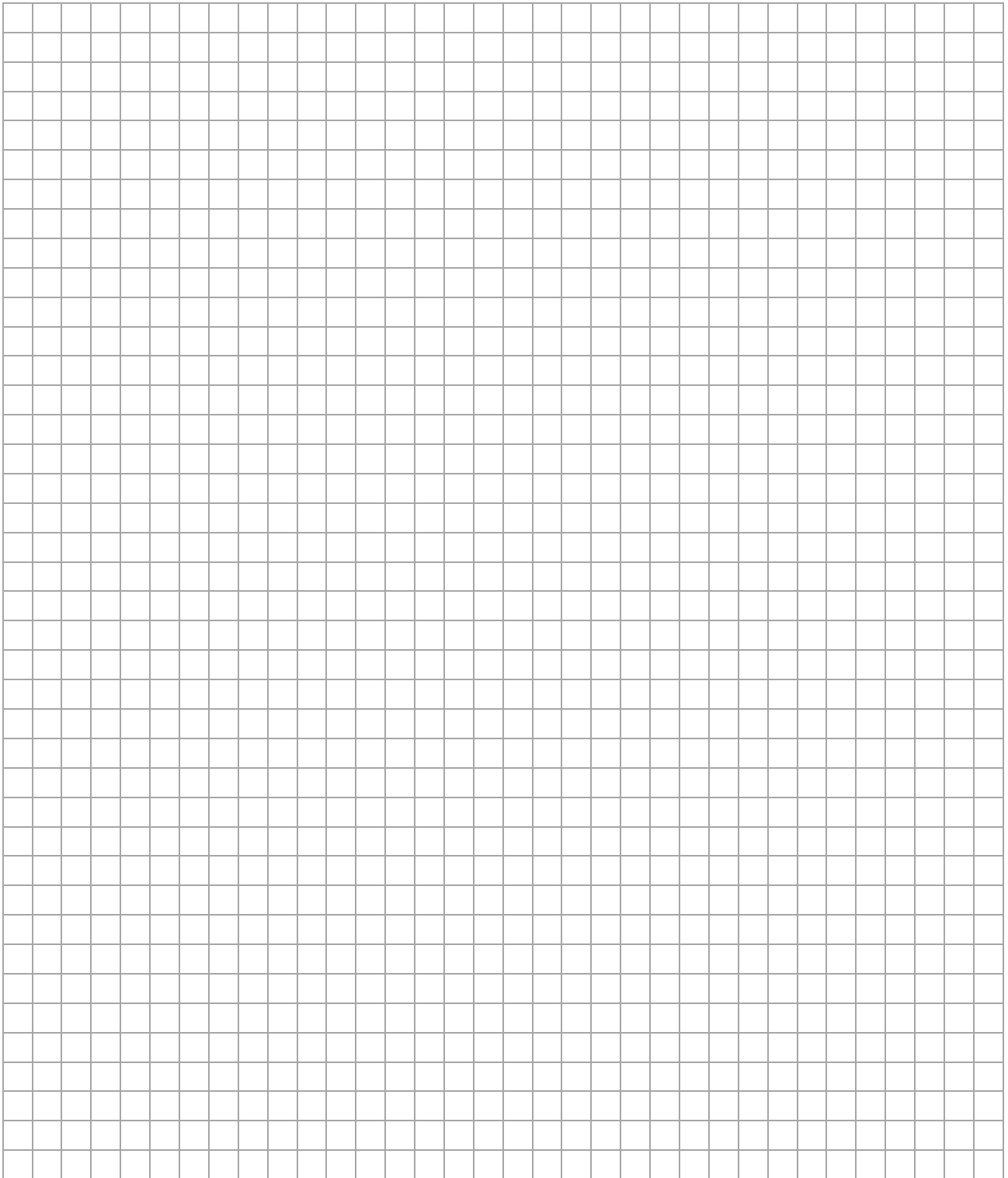
- (d) The line k passes through the point $(-3, 2)$ and is perpendicular to the line l .
Find the equation of the line k in the form $ax + by + c = 0$.

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You may use this page for extra work.



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Pre-Junior Certificate, 2017 – Higher Level

Mathematics – Paper 2

Time: 2 hours, 30 minutes

